PixelRoot baseline (evaluated)
upgrade path (M1–M3)
Per-frame gain
Global analog/digital gain
(every sensor today)
cost: trivial (firmware)
Per-column gain
Column-parallel PGA or
ADC reference trim
cost: low (existing column ADC)
Per-region gain
Tile-programmable gain via
row/column gating
cost: moderate (control + SRAM)
Per-pixel gain
M1 conversion-gain select,
M2 coded exposure,
M3 stacked pixel ADC
cost: higher (area, pitch, yield)
finer spatial resolution of the embedded mark · more manufacturing investment